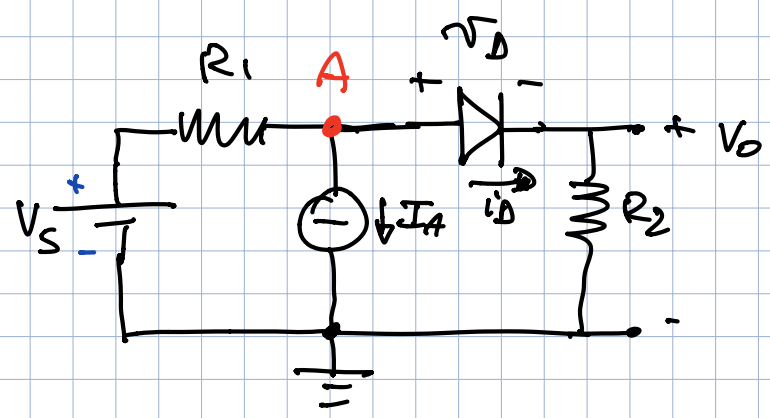


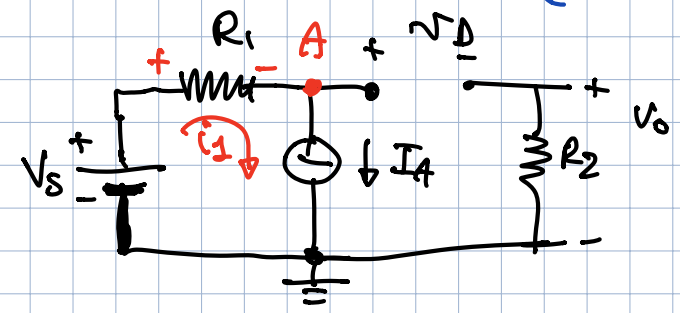
$R_1 = R_2 = 1\,K\Omega$

$I_A = 1\,mA$

$V_S = 6\,V$

$V_{ON} = 1\,V$

HP DIODO "OFF" ($v_D < V_{ON}$)

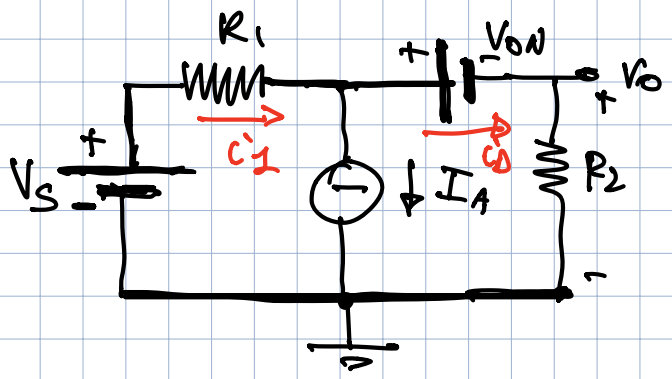


$V_A = V_S - R_1 I_A = 6\,V - 1\,mA \cdot 1\,K\Omega = 5\,V$

$v_D = V_A - V_O = 5\,V - 0 = 5\,V$

$v_D > V_{ON} \Rightarrow$ DIODO ON!

CONSIDERO DIODO "ON"



$V_S = R_1 i_1 + V_{ON} + R_2 i_D$

$i_D = i_1 - I_A$

$V_S = R_1 i_1 + V_{ON} + R_2 i_1 - R_2 I_A$

$\Rightarrow i_1 = \dfrac{V_S - V_{ON} + R_2 I_A}{R_1 + R_2} = \dfrac{6V - 1V + 1V}{2\,K\Omega} = 3\,mA$

$i_D = i_1 - I_A = 3\,mA - 1\,mA = 2\,mA$

$i_D > 0 \Rightarrow$ OK DIODO "ON"

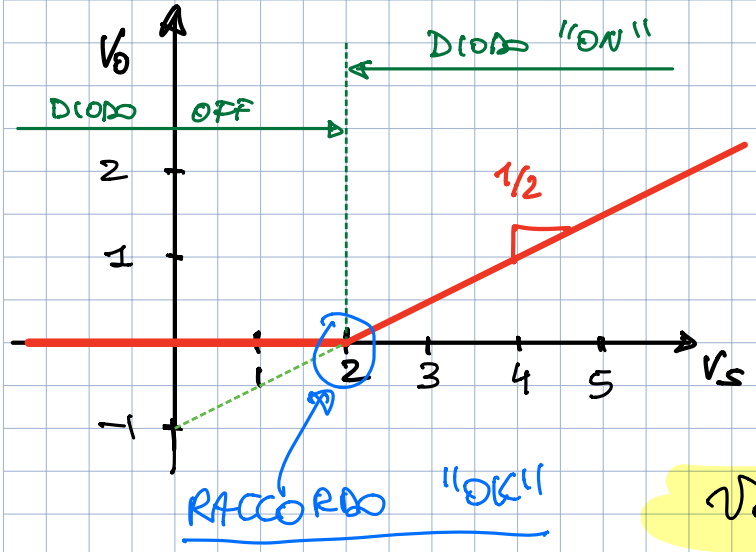
$v_O = i_D R_2 = 2\,V$

FACCIAMO V_S VARIABILE $\Rightarrow$ CONSIDERO DIODO "OFF"

$v_D = V_A = V_S - R_1 I_A < V_{ON}$

$\Rightarrow V_S < V_{ON} + R_1 I_A = 1\,V + 1\,V = 2\,V$!

CON $V_S < 2V$ DIODO = OFF e $V_O = 0$!



Con $V_S > 2V \Longrightarrow$ DIODO "ON"

$$i_D = i_1 - I_A = \frac{V_S - V_{ON} + R_2 I_A}{R_1 + R_2} - I_A$$

$$v_0 = R_2 i_D$$

$$= \frac{R_2}{R_1+R_2} V_S + \frac{R_2}{R_1+R_2}\underbrace{(-V_{ON} + R_2 I_A)}_{-1+1=0} - I_A R_2$$

$$v_0 = \frac{1}{2} V_S - 1\ V \qquad (V_S > 2V)$$

$v_0(2V) = 0V$ OK RACCORDO